Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the stress in the wire.

- $A = \pi \times 2^2 = 12.6$ mm²
- $\sigma = 300 \div 12.6 = 24$ N/mm²

2 The stress in a steel wire carrying 8.0 kN must not exceed 200 N/mm². Calculate the minimum diameter of wire needed.

- $A = F \div \sigma = 8000 \div 200 = 40$ mm²
- $d = \sqrt{(4 \times 40 \div \pi)} = 7.1$ mm (round up for safety)

3 A roof-support column has a cross-section of 0.73 m². The maximum working stress is 4.0 MN/m². Calculate the maximum compressive load.

- $F = \sigma \times A = 4\,000\,000 \times 0.73$
- F = 2 920 000 N (2.9 MN)

4 A North Sea oil platform has three legs, each 15.0 m in diameter, sharing a 120 MN load equally. Calculate the stress in each leg.

- Force per leg = 120 MN ÷ 3 = 40 000 000 N
- $A = \pi \times 7.5^2 = 176.7$ m²
- $\sigma = 40\,000\,000 \div 176.7 = 226\,000$ N/m² (0.23 N/mm²)